

Bioinformatics Bootcamp 2026

Assessment #01

Question: 01

Answer: I would recommend 2 databases:

i- NCBI Genbank

ii- ClinVar

i- It is a massive global library of all DNA sequences. The researcher can paste their sequence into a tool called "Blast" to search Genbank. If a sequence appears, it means someone else saw it first. It tells gene name, the organism of which the sequence belongs and any previous literature about it.

ii- The researcher suspects a link to a genetic disorder. This database specifically catalogs human genetic variations and connects them to actual health conditions.

DATE: ___/___/20___

M T W T F

By checking these databases, the researcher can find out if this specific mutation had been blamed for disorder, saving them months of lab work.

DATE: ___/___/20

(M) (T) (W) (T) (F)

Question: 02

Answer: I recommend NCBI genome and the PATRIC (BV-BRC) database.

i- NCBI Genome: This provides the complete genetic blueprint for thousands of organisms. The scientists can download the full genomes of related bacterial species. They use BLAST to line up the genome side by side.

ii- BV-BRC: It is specifically built for bacteria and viruses. It has tools "Proteome Comparison" tool which genes are shared across different bacterial strains.

1
These databases will highlight the 'conserved' genes. Because these genes are stubbornly kept by evolution, they are critical for bacteria's survival and make perfect targets for new antibiotics.

Question: 03

Answer: The brand should look at UniProt and NCBI protein.

i- This is standard for protein information. The data is verified and reliable. If company's protein matches something in Swiss-Prot

they will learn about its three dimensional structure, its function, and what species express it.

ii- NCBI Protein: This is a broader repository that includes automated translations of DNA sequences from GenBank.

By running a BLASTp search against these databases, the company will find "homologs". If a similar protein exists in a well-studied baker's yeast or bacterium, the company can infer how their new micro-organism's proteins behaves.

Question: 04

Answer: I would recommend:

i- InterPro.

ii- PDB (Protein Bank Data).

i- Proteins are often made of smaller functional building blocks called "domains" or "motifs". Even if the whole protein is unknown, InterPro scans the sequence for these recognizable pockets. It combines

data from multiple databases like Pfam and Prosite to say that this specific region looks exactly like a binding site for sugar.

ii- If protein has a known 3D structure, PDB provides the exact physical shape.

InterPro will tell the scientist the protein's "signature" moves. Understanding these domains allows the scientist to confidently predict the protein's biological job in the cell.

Question: 05

Answer:

Step # 01:

If it contains 'U' it's RNA. If it mostly contains A, T, C, G it's DNA. If it uses a wide variety of letters like M, W, L, P it's a protein.

DATE: ___/___/20___

M T W T F S S

Step #02

Go to NCBI and BLAST suite. If Step 1 suggested it was DNA/RNA use BLASTn, if it looked protein, use BLASTp. If we are unsure then use blastx which translates a nucleotide sequence into all possible protein variations and checks them.

Step #03:

Look at the top hits in BLAST results. The database will reveal the exact organism the sequence belongs to, the gene name, its percentage of identity.

Step #04:

Once BLAST gives you a name, cross-reference it with UniProt or Ensembl to read its full bio, confirming exactly what your sequence is and does.